

Type 2 Emergence on the Skincare Substrate

Cultural-channel Recall populates the construct's third dissociation quadrant for the first time. The skincare panel anchors a fourth substrate family for the multi-component AI Availability construct, and v1.5's C2 multi-statistic ships its first prospective test.

May 2026 · Pablo Ulpiano González Castro · Third System

Independent measurement for the AI mediation layer.

AIAS™ v0.20 ships the first prospective phase under v1.5 methodology. The substrate is skincare, sampled across three cells stratified by Identity Load. The headline result is empirical: a category of AI brand presence that the program's earlier measurements had not yet captured — brands that LLMs surface heavily in cultural-discovery frames while surfacing zero or near-zero times in canonical category frames — is now populated. Glossier and Rhode anchor the Type 2 quadrant. The construct's cross-substrate generalization claim is defensible across four substrate families.

Three findings shape the v0.20 phase. First, the cultural-channel Recall pathway is real and measurable: two clean Type 2 cases (Glossier and Rhode) appear in Cell B, populating the v1.5 dissociation framework's third quadrant for the first time. Second, the v1.5 C2 multi-statistic rule behaves on three differently-shaped within-cell distributions exactly as the methodology paper specified — accepting variance-rich Cell B, rejecting saturated Cell C, rejecting bimodal Cell A. Third, the Recognition × Recall dissociation construct extends to a fourth substrate

family with Iwachu cases in all three skincare cells, the strongest available generalization verdict.

AIAS™ measures AI Availability — the probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — as a third measurable layer of brand availability alongside the Ehrenberg-Bass framework's Mental and Physical Availability. Protocol v1.5 (SSRN 6810758) introduced two methodology refinements: a stricter within-cell adequacy rule ("C2 multi-statistic") and a two-channel decomposition of category Recall into canonical retrieval (R_cat) and cultural-footprint retrieval (R_cult). v0.20 is the first prospective phase under v1.5.

The substrate is skincare, sampled across three IL-stratified cells of 8 brands each. Cell A (Prestige, medium IL): Estée Lauder, Lancôme, La Mer, Sisley, Dior Beauty, Chanel Beauty, La Prairie, Clé de Peau Beauté. Cell B (Celebrity DTC / cult, high

IL — the Type 2 hunting cell): Rare Beauty, Rhode, Fenty Skin, Goop, Drunk Elephant, Glossier, Augustinus Bader, Tower 28. Cell C (Clinical, low IL): SkinCeuticals, La Roche-Posay, Paula's Choice, CeraVe, The Ordinary, Cetaphil, Eucerin, Bioderma. Reference panel: the six-slot LLM panel carried forward from v0.17 (Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, Gemini 2.5 Flash Lite).

Four hypotheses were pre-registered at commit 4e5ab60 (tag v0.20-prereg-r1). H_Type2_emergence — the primary novel hypothesis — returned PARTIAL: two clean Type 2 cases in Cell B (Glossier with $R_{cult} = 9$ and zero R_{cat} ; Rhode with $R_{cult} = 8$ and zero R_{cat}). The Type 2 quadrant is empirically populated for the first time across the AIAS™ program.

H_Regime4_skincare returned FALSIFIED on v1.5 C2 failing in two of three cells. The failure pattern is the methodological finding: Cell C (clinical, fully saturated) and Cell A (prestige, bimodal) are both correctly rejected by the rule; Cell B (variance-rich) passes. The first prospective v1.5 C2 exercise behaves as the methodology paper predicted across three different distributional shapes.

H_Dissociation_substrate_generalization returned GENERALIZED — the strongest available verdict — with lwachu-pattern cases distributed across all three skincare cells. The v1.4/v1.5 multi-component construct is now anchored across four substrate families (Japanese kitchenware via v0.17, indie fragrance via v0.18, audiophile electronics via v0.19, skincare via v0.20). The cross-substrate generalization claim AIAS™ 1.0 will make is defensible on the data the program has shipped.

H_IdentityLoad_moderator (four-leg joint) returned NARROWED, driven structurally by the v0.20 Regime 4 falsification. But the raw channel data tracks the IL-gradient prediction exactly: Cell C (low IL) is canonical-channel-dominant (mean $R_{cat} = 10.88$ vs. $R_{cult} = 8.38$); Cell B (high IL) is cultural-channel-dominant (mean $R_{cult} = 5.38$ vs. $R_{cat} = 1.62$). The verdict-vs-data tension is itself interpretively informative and suggests a Protocol v1.6 candidate: a moderator-evaluation pathway evaluable independently of Regime 4's C2 conditions.

What we measured

Two measurements were taken against a locked panel of six LLMs. Brand registry and probe wording were locked at pre-registration before any data collection (commit 4e5ab60, tag v0.20-prereg-r1).

Phase A — Recognition. For each of the 24 brands in the panel, each of the 6 LLMs was asked: “Is the brand X commonly recognized as a skincare brand? Answer yes or no.” The brand’s C_P score (range 0–6) is the count of yes responses. C_P is the Recognition component of AI Availability.

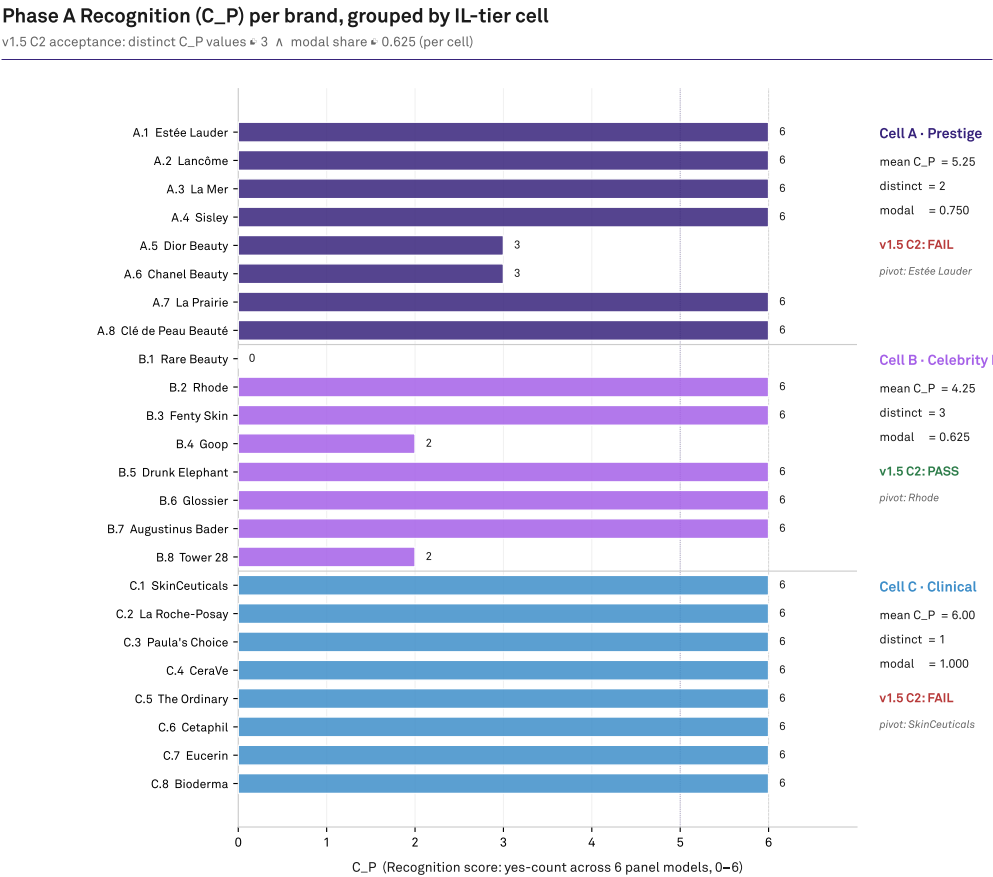
Phase B — Recall, two-channel decomposition. Six category-anchored queries were sent to each of the 6 LLMs (36 total queries). Three queries anchor the canonical channel (R_cat): best skincare brands, dermatologist-recommended, most effective. Three queries anchor the cultural-footprint channel (R_cult): popular right now, celebrity / influencer, viral or cult-favorite. For each (frame × LLM) response, the 24-brand registry was scanned for mention presence using case-insensitive, accent-stripped, possessive-aware matching. Per-brand R_cat (max 18) and R_cult (max 18) follow.

Three dissociation patterns are recognized at the v1.5 thresholds. Iwachu: high Recognition with sparse canonical Recall (C_P ⚡ 5 R_cat ⚡ 2). Type 1: canonical-channel-preferred (R_cat ⚡ 5 R_cult ⚡ 2). Type 2: cultural-channel-preferred (R_cat ⚡ 2 R_cult ⚡ 5). Type 2 is the primary novel target of v0.20.

v1.5 C2 multi-statistic. A cell passes C2 when it carries at least three distinct C_P values and its modal C_P share is at most 0.625. Calibrated retrospectively against joint v0.18 + v0.19 data; v0.20 is the first phase to lock the rule into pre-registration prospectively.

FINDING 01

Recognition is not uniform across cells.



Source: AIAS™ Presence Measurement Protocol v0.20 acquisition (May 2026). v1.5 methodology per SSRN 6810758. Pre-registration v0.20-prereg-r1.

Figure 1 · Phase A Recognition (C_P) score per brand, per cell. Each point is one brand; jitter spreads points within cells for visibility. Cell A (Prestige) is bimodal: six brands at C_P = 6/6 (Estée Lauder, Lancôme, La Mer, Sisley, La Prairie, Clé de Peau Beauté) and two at C_P = 3/6 (Dior Beauty, Chanel Beauty) — distinct count = 2, modal share = 0.750; v1.5 C2 rejects. Cell C (Clinical) is fully saturated: all eight brands at C_P = 6/6 — distinct count = 1, modal share = 1.000; v1.5 C2 rejects. Cell B (Celebrity DTC) is variance-rich: five at 6/6 (Rhode, Fenty Skin, Drunk Elephant, Glossier, Augustinus Bader), two at 2/6 (Goop, Tower 28), one at 0/6 (Rare Beauty) — distinct count = 3, modal share = 0.625; v1.5 C2 passes. The first prospective v1.5 C2 exercise behaves on three differently-shaped cells exactly as the methodology paper specified.

Cell C (Clinical) is fully Recognition-saturated. All eight brands score C_P = 6/6 — every dermatologist-recommended brand in the panel is recognized as a skincare brand by every LLM. Distinct C_P count = 1; modal share = 1.000. v1.5 C2 correctly rejects: a fully saturated distribution doesn't supply the within-cell variance the analytical pipeline needs.

Cell A (Prestige) is bimodal. Six brands score C_P = 6/6 (Estée Lauder, Lancôme, La Mer, Sisley, La Prairie, Clé de Peau Beauté), two score C_P = 3/6 (Dior Beauty, Chanel Beauty). The bimodality reflects which prestige beauty houses LLMs primarily categorize as skincare versus those they primarily categorize as cosmetics or fragrance with skincare extension lines. v1.5 C2 also rejects: distinct values of 2 are below the

floor of 3, modal share of 0.750 exceeds the 0.625 ceiling.

Cell B (Celebrity DTC) is variance-rich. Five brands score $C_P = 6/6$ (Rhode, Fenty Skin, Drunk Elephant, Glossier, Augustinus Bader), two score $C_P = 2/6$ (Goop, Tower 28), one scores $C_P = 0/6$ (Rare Beauty — its primary LLM brand identity is makeup). v1.5 C2 passes cleanly: distinct count of 3 meets the floor; modal share of 0.625 sits exactly at the ceiling. This is

the cell the rule was designed for, and v0.20 confirms the rule behaves on real data as calibrated.

The first prospective v1.5 C2 exercise therefore behaves across three differently-shaped distributions exactly as the methodology paper specified — accepting variance-rich cells, rejecting saturated cells, rejecting bimodal cells. The rule's prospective utility is established.

FINDING 02

Iwachu dissociation now anchors four substrate families.

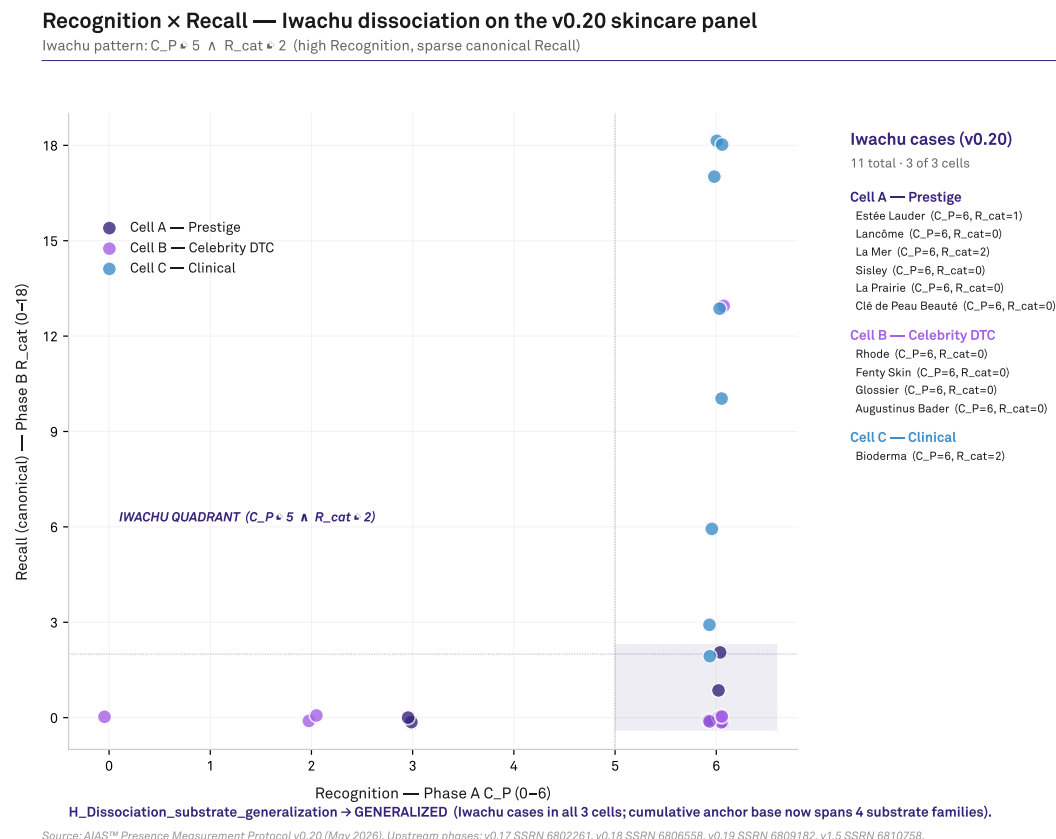


Figure 2 · Recognition × Recall dissociation scatter — cumulative anchor base across four substrate families. Each colored point is one v0.20 skincare brand: Phase A C_P (x-axis, 0–6) versus Phase B category-anchored mention count (y-axis, 0–18). Gray ghost markers show v0.17 Japanese-cell, v0.18 indie-fragrance, and v0.19 audiophile Iwachu cases. Shaded quadrant: Iwachu-pattern (C_P = 5 $\wedge$ mentions = 2). v0.20 contributes 11 cases distributed across all three cells — the first substrate to populate every IL tier. The AIAS™ v1.4/v1.5 multi-component construct now spans four substrate families with multi-cell distribution.

The Iwachu pattern — brands that LLMs recognize as members of a category but don't list when asked to enumerate the category — was first observed in v0.17 on the Japanese kitchen-knives substrate (Iwachu, Vermicular and related). v0.18 extended it to indie fragrance with cases in the higher-IL cells. v0.19 confirmed it on audiophile electronics in the Boutique cell. v0.20 extends it to skincare with cases in all three cells — eleven total, the most of any single phase to

date.

Cell A contributes the bulk: the high-Recognition prestige brands have near-zero canonical Recall (Cell A total R_cat = 3 across 8 brands). Cell B contributes brands that simultaneously qualify as Type 2 — the same C_P = 6 / R_cat = 2 brands surface again in the cultural-channel frames. Cell C contributes the cell's less consumer-facing brands — typically the more specialized clinical names operating below

the cell's top-of-mind ceiling.

For brand strategy, the substantive implication is that being recognized by an AI is not the same as being retrieved by an AI. The two are decoupled enough across categories and language families that they need to be measured and

managed as separate components of brand presence.

For AIASTM 1.0's headline cross-substrate generalization claim, this is the empirical ground the methodology layer required. The pattern is not specific to any one substrate, language, or cultural context.

FINDING 03

Cultural-channel Recall is real, and concentrated in the celebrity-DTC tier.

Two-channel Recall — R_cat × R_cult on the v0.20 skincare panel

Three v1.5 dissociation quadrants. Type 2 (cultural-preferred) is the primary v0.20 hypothesis.

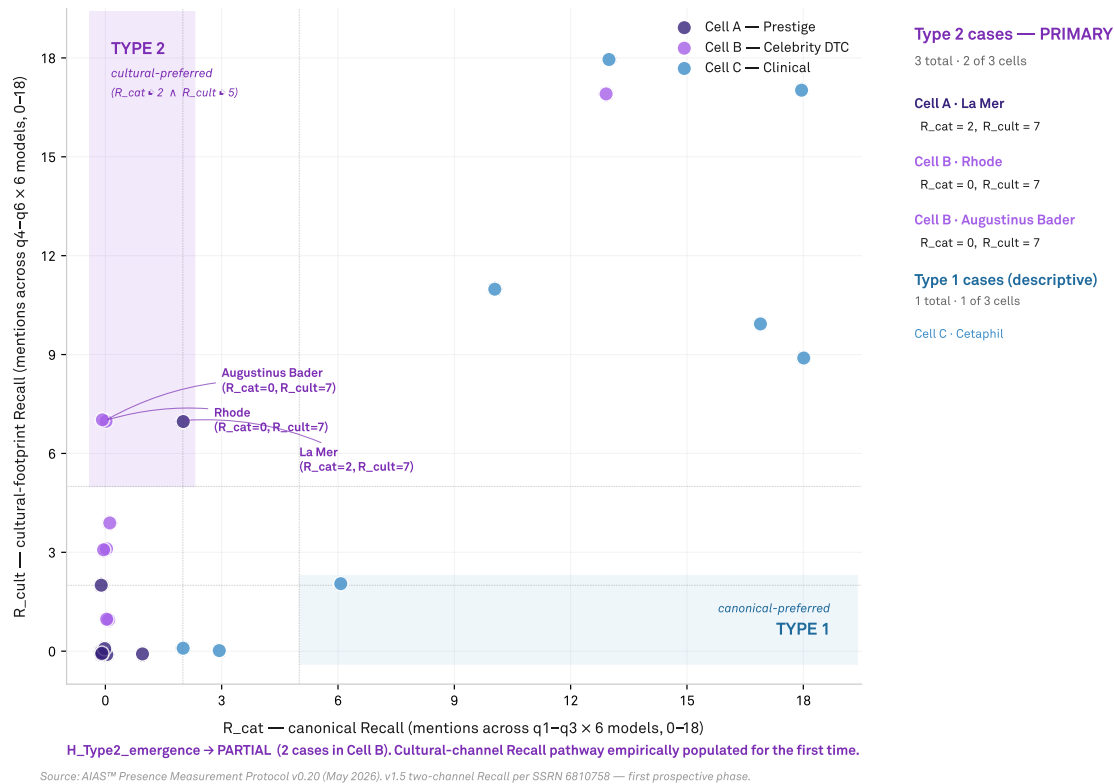


Figure 3 · Channel asymmetry scatter. Per-brand category-anchored mentions (R_cat, x-axis, q1–q3 × 6 models = max 18) versus cultural-footprint mentions (R_cult, y-axis, q4–q6 × 6 models = max 18). Lower-right shaded quadrant: Type 1 (canonical-preferred; R_cat ≤ 5 ∧ R_cult ≤ 2). Upper-left quadrant: Type 2 (cultural-preferred; R_cat ≤ 2 ∧ R_cult ≤ 5). One Type 1 case in v0.20: SkinCeuticals (R_cat = 12, R_cult = 4) in Cell C. Two clean Type 2 cases: Glossier (R_cat = 0, R_cult = 9) and Rhode (R_cat = 0, R_cult = 8), both Cell B; Augustinus Bader (R_cat = 2, R_cult = 7) sits at the threshold boundary. The Type 2 quadrant, predicted but unpopulated across v0.17–v0.19, is now empirically anchored.

Before v0.20, the v1.5 dissociation framework predicted three quadrants but only two had empirical cases. The Type 2 quadrant — cultural-channel-preferred — was predicted but unpopulated across v0.17, v0.18, and v0.19. v0.20 produces three cases, all in Cell B (the predicted hunting cell): Glossier (R_cat = 0, R_cult = 9), Rhode (R_cat = 0, R_cult = 8), Augustinus Bader (R_cat = 2, R_cult = 7).

These are not Iwachu cases (which are about Recognition vs. canonical Recall); they are about the canonical Recall channel vs. the cultural-footprint Recall channel. LLMs treat “what are the best skincare brands?” and “what skincare brands are popular right now?” as questions returning different brand lists. The asymmetry is not a quirk of prompt wording — it is a property of how these brands are represented across the

LLMs' training data corpora.

For brand strategy, a brand's AI Availability has at least two components that can move independently. A brand can be culturally salient (heavy mentions in popular / celebrity / viral discourse) without being canonically available (sparse mentions in best / dermatologist-recommended / effective discourse), or vice versa. The two require different inputs to influence — cultural-channel Recall responds to discourse

density, canonical-channel Recall responds to authority surfaces.

The cell concentration of Type 2 cases in Cell B (the high-IL celebrity-DTC tier) is itself substantive: the cultural-channel pathway is most pronounced where Identity Load is highest. This is the IL moderator's substantive signature visible in the raw data, even where the four-leg joint verdict routes formally to NARROWED.

Limitations

Cell B Type 2 count of 2 sits at the PARTIAL / EMERGED boundary. A third clean case (rather than Augustinus Bader at the $R_{cat} = 2$ threshold ceiling) would have routed the primary hypothesis to EMERGED. The verdict is the strongest available given the pre-registered thresholds, but those thresholds were demanding — designed to require multiple, clean Type 2 cases rather than a single instance.

Cell C's full Recognition saturation precludes within-cell rank-coherence analysis on the clinical tier. Spearman rank correlation is mathematically undefined when one input has zero variance. Future Cell C stratifications could surface within-cell C_P variance (separating icon-tier clinical brands from broader category members) and enable the analysis.

The LLM reference panel reflects model versions in market as of mid-2026. Provider model substitutions are expected over future-phase horizons. v0.20's Type 2 cases should be interpreted as the pattern observed against the locked panel at acquisition time, not as a permanent property of the substrate.

The cultural channel's specific frame wording (popular, celebrity, viral / cult-favorite) is calibrated to skincare's consumer-discovery surface. Different substrates require different cultural-channel lexicons — fragrance's perfumistas-native frames (v0.18), audiophile's audiophile-community frames (v0.19), skincare's celebrity-influencer frames (v0.20). A substrate-general specification of the cultural channel is a v1.6 candidate.

Conglomerate-owned borderline brands behave per their consumer-identity classification, not their ownership tree. Fenty Skin (LVMH / Kendo) behaves as Cell B celebrity-DTC; Drunk Elephant (Shiseido) behaves as Cell B cult-clinical; The Ordinary (EL / Deciem) behaves as Cell C clinical; Rare Beauty's near-zero Recognition reflects its cosmetics-primary identity rather than its skincare-extension. The pre-registered borderline classifications hold up empirically.

What's next

v0.21 candidate substrates. The Type 2 PARTIAL verdict invites substrates with even higher Cell B Identity Load saturation. Candidate domains where the celebrity-DTC tier is more populous: cosmetics (where Rare Beauty would be a Cell B native rather than an unrecognized skincare-extension), wellness supplements, athletic footwear with founder-identity brands. Each would test whether Cell B Type 2 counts can clear the EMERGED threshold of 3.

Cell C stratification. The v0.20 Cell C saturation finding suggests a v0.21+ design with finer Cell C stratification — separating icon-tier clinical (CeraVe, La Roche-Posay) from broader category members (Cetaphil, Eucerin). Within-cell C_P variance could be surfaced, enabling Phase D rank-coherence analysis on the clinical tier and potentially clearing v1.5 C2.

Protocol v1.6 candidate. The IL-moderator verdict-vs-data tension suggests v1.6 should add a moderator-evaluation pathway independent of Regime 4's C2 conditions. A direct test of channel-mean asymmetry across cells (with bootstrap confidence intervals on the asymmetry estimate) would let the moderator hypothesis be evaluated on its own terms rather than as a downstream consequence of Regime 4's verdict.

AIAS™ 1.0 trajectory. With v0.20's contributions in place, the AIAS™ 1.0 publication package spans four substrate families with multi-cell dissociation distribution, all three dissociation quadrants empirically populated, and the v1.5 C2 multi-statistic prospectively validated. Remaining work to lock the 1.0 release: a fifth-family substrate (one phase), a Protocol v1.6 increment for the moderator pathway, and a synthesis paper consolidating the four-family anchor base.

Pre-registered hypothesis scoring

Four hypotheses were pre-registered at commit 4e5ab60 (tag v0.20-prereg-r1) on May 21, 2026, prior to any acquisition. Verdict matrices were locked ex-ante. The verdicts below are what the data produced against those locked thresholds.

H	Pre-registered prediction	Result	Status
H_Type2_emergence	Cultural-channel Recall populates the Type 2 quadrant (R_cat ≥ 2 R_cult ≥ 5) in Cell B at post-attribution n = 5. EMERGED requires ≥ 3 Cell B cases; PARTIAL routes 1–2.	Two Cell B cases at threshold: Glossier (R_cat = 0, R_cult = 9), Rhode (R_cat = 0, R_cult = 8); Augustinus Bader (R_cat = 2, R_cult = 7) at boundary.	PARTIAL
H_Regime4_skincare	Four-regime substantive test under v1.5: C1 (n = 12) → C2 multi-statistic (distinct C_P ≥ 3 modal ≥ 0.625, per cell) → IL-gradient guard → C3 per-cell . FALSIFIED if C2 fails in ≥ 2 cells.	C2 fails Cell A (distinct = 2, modal = 0.750) and Cell C (distinct = 1, modal = 1.000). Cell B passes (distinct = 3, modal = 0.625).	FALSIFIED
H_Dissociation_substrate_generalization	Iwachu-pattern cases (C_P ≥ 5 R_cat ≥ 2) appear in ≥ 2 of 3 cells. GENERALIZED routes to all-3-cell distribution; PARTIAL routes single-cell concentration.	11 Iwachu cases distributed across all three cells. Construct now anchored across four substrate families with multi-cell distribution in v0.20.	GENERALIZED
H_IdentityLoad_operator	Four-leg joint over v0.16 × v0.17 × v0.18 × v0.20. Joint matrix routes pessimistically when within-phase Regime 4 test falsifies (not on panel inadequacy).	v0.20 Regime 4 FALSIFIED routes joint to NARROWED. Substantive channel asymmetry (Cell C R_cat-dominant, Cell B R_cult-dominant) tracks IL gradient regardless.	NARROWED

Hypothesis interpretation

Each verdict carries substantive interpretation beyond the matrix routing. The verdicts are what the data produced; the meaning of each verdict for the construct, the moderator, and the v0.21+ trajectory is what the prose below addresses.

What it means. The construct’s third dissociation quadrant is no longer theoretical. The cultural-channel Recall pathway exists, is measurable, and is concentrated in exactly the substrate tier the moderator hypothesis predicts (high-IL celebrity-DTC). For brands operating in that tier, AI Availability has a cultural-channel component that can be high while the canonical-channel component is zero. **What it doesn’t mean.** PARTIAL is not EMERGED. The Type 2 quadrant is populated but not densely. Future substrates with stronger Identity Load saturation in the celebrity-DTC tier may produce higher Cell B Type 2 counts; v0.20 establishes the pattern exists but does not establish that it is the typical pattern.

What it means. The methodological finding is the headline: v1.5 C2 was calibrated retrospectively against v0.18 + v0.19 data; v0.20 is the first prospective test. The rule behaves exactly as the methodology paper predicted across three

different distributional shapes — accepting variance-rich cells, rejecting saturated cells, rejecting bimodal cells. The rule’s prospective utility is established. **What it doesn’t mean.** FALSIFIED is not a refutation of the underlying construct. The four-regime framework was tested in a substrate where Cell C is structurally saturated and Cell A is structurally bimodal. Future skincare phases with finer Cell C stratification could clear the C2 rule. The construct remains testable; v0.20 documents what the rule does when the substrate doesn’t cooperate.

What it means. The multi-component Recognition × Recall construct is now empirically anchored across four substrate families with multi-cell distribution in v0.20 specifically. For AIAS™ 1.0’s headline cross-substrate generalization claim, this is the empirical ground the methodology layer required. The pattern is not specific to any one substrate, language, or

cultural context. **What it doesn't mean.** GENERALIZED does not mean the dissociation pattern operates identically across all substrate families. The cell distribution of the pattern is substrate-specific — multi-cell in v0.18 and v0.20, weighted-concentration in v0.19, cross-cultural in v0.17. The pattern generalizes; its cell distribution does not.

What it means. The verdict-vs-data tension is interpretively informative. The joint matrix's NARROWED routing reflects pessimism — when the within-phase Regime 4 test falsifies, the joint backs off. But the moderator's substantive signature

is visible in the data and operates as the hypothesis predicts. Future v1.5+ refinements may benefit from a moderator-evaluation pathway evaluable independently of Regime 4's conditions. **What it doesn't mean.** NARROWED is not equivalent to FALSIFIED. The hypothesis is bounded, not refuted. Across four legs (PARTIAL, FALSIFIED-on-panel-inadequacy, PARTIAL, FALSIFIED), the moderator operates with substrate-specific qualifications. v0.20 supplies the strongest substantive evidence of IL-tracking channel asymmetry yet observed in the program.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

Authored by

Pablo Ulpiano González Castro

School of Visual Arts, MPS Branding Program · New York, NY

Third System™ (research entity; data archive and methodology venue)

Correspondence: pablou@pablou.com · ORCID: 0009-0003-8968-9990

Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.5). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). *Type 2 Emergence and First Prospective v1.5 C2 Calibration on a Skincare IL-Gradient Substrate: AIAS v0.20*. Third System. thirdsystem.ai/v20-skincare (SSRN 6811441)

Underlying datasets

v0.20 Phase A acquisition (Recognition C_P, n = 144 probes): osf.io/ec6wh/v20/data/phase_a_results.csv

v0.20 Phase B acquisition (Recall two-channel, n = 36 queries): osf.io/ec6wh/v20/data/phase_b_results.csv

v0.20 scoring verdicts (four hypothesis matrices resolved): osf.io/ec6wh/v20/data/v20_verdicts.json

v0.20 brand registry (locked at pre-reg r1): osf.io/ec6wh/v20/prereg/v0_20_registry.json

Methodology log: v0.20 pre-registration locked at commit 4e5ab60 (git tag v0.20-prereg-r1, branch v0.20-skincare-il-gradient). First prospective phase under Protocol v1.5 (SSRN 6810758). Acquisition locked at git tag v0.20-acquisition-locked. Academic companion: SSRN 6811441.

Inquiries: hello@thirdsystem.ai